Lesson notes · for you

How AI sees

AI · Level 1 · Lesson 3

Time	Materials
27 min	Squared paper, or a 10 × 10 grid drawn with a ruler; A pencil; A friend or family member

Learning goals

- Explain that a picture is a grid of numbers called pixels.
- Explain how vision AI learns patterns in those numbers from labelled photos.
- Give reasons vision AI can be fooled: light, angle and things it never saw.

Activity: Picture in numbers

1. Draw (5 min): Draw a 10 × 10 grid on squared paper (or with a ruler) and shade a simple picture: a house, a ball, a smiling face or the first letter of your name.
2. Encode (7 min): On a second grid, write a number in every square: 0 for white, 9 for black, and 3 to 6 for grey. Don’t show anyone the shaded picture.
3. Swap (6 min): Swap number grids with a partner. Each of you shades an empty grid using only the numbers: darker for bigger numbers.
4. Compare (4 min): Put the original and the copy side by side. Did the picture come back? Which squares were hardest, and why?
5. Fool it (5 min): Add 3 to every number to “dim the light”, and copy the grid again. Talk about it: the picture is the same, but every number changed. What would a vision AI that only learned bright photos do?

Each person needs two or three 10 × 10 grids and a pencil. Keep numbers between 0 and 9: when “dimming the light”, stop at 9.

Discussion

- What would a vision app need to learn before it could recognise a fruit from your region?
- When has your phone’s camera or face unlock got something wrong?
- Why should a person check a vision AI’s answer when it really matters?

Check yourself

- Write a 3 × 3 picture as numbers, and give one reason vision AI can be fooled.

Try it this week

Draw a simple picture on squared paper, write it as numbers (0 for white, 9 for black), and give only the numbers to someone. Can they draw your picture back?

Worksheet

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Name _____ Class _____ Date _____

1 **Face unlock works every morning in your room, but keeps failing in bright sun outside. Why?**

◯ The light changed the numbers ◯ The camera can't see in sunlight ◯ It forgot your face

2 **When a phone looks at a photo of a mango, it sees a mango, just as you do.**

◯ True ◯ False

3 **Complete the sentence.**

A picture is a _____ of tiny squares called _____, and each one is stored as a _____.

Word bank: colour · name · pixels · grid · number

4 **Match each problem to why vision AI gets confused.**

A photo taken in dim light	• • • •	The shapes look different
A leaf photographed from the side	• • • •	Every pixel number changes
A fruit it never saw in training	• • • •	Too few clear pixels to show the pattern
A tiny, blurry photo	• • • •	It guesses the nearest thing it knows

✎ **Think of a photo your phone got wrong, or one that might fool it. What changed the numbers: the light, the angle, or something it never saw?**

Check your answers

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1 **The light changed the numbers**

Bright sun changes every pixel's number and throws hard shadows and glare across your face, so the pattern no longer matches the one it learned.

2 **False**

It gets only a grid of numbers. "Mango" is a label it predicts from patterns in those numbers, learned from many labelled photos, and it can be wrong.

3 **A picture is a grid of tiny squares called pixels, and each one is stored as a number.**

A picture is a grid of pixels, and each pixel is stored as a number.

4 **A photo taken in dim light — Every pixel number changes**
A leaf photographed from the side — The shapes look different
A fruit it never saw in training — It guesses the nearest thing it knows
A tiny, blurry photo — Too few clear pixels to show the pattern

Each problem changes the numbers, the shapes or what the AI has seen before.